

1. Administrative & Study Identification

Full Study Title: PHASE 1/2A DOSE ESCALATION AND EXPANSION STUDY EVALUATING SAFETY, TOLERABILITY, PHARMACOKINETIC, PHARMACODYNAMICS AND ANTI-TUMOR ACTIVITY OF PF-06873600 AS A SINGLE AGENT AND IN COMBINATION WITH ENDOCRINE THERAPY **Short Title/Acronym:** A Study of PF-06873600 in People With Cancer **Protocol Number:** C3661001 **NCT Number:** NCT03519178 **Sponsor Name:** Pfizer Inc. **Investigational Product:** PF-06873600 (Selective CDK2/4/6 Inhibitor) **Phase of Development:** Phase I/IIa **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate safety, including dose-limiting toxicities (DLTs) in the first 28-day cycle, adverse events (AEs), and laboratory abnormalities; and to select the recommended dose for expansion (RDE) for PF-06873600 as a single agent and in combination with endocrine therapy. The optimization target is to achieve objective tumor response and prolonged disease control. **Secondary Objectives:** To evaluate pharmacokinetics (PK), pharmacodynamics (PD), and preliminary anti-tumor activity (including Objective Response Rate [ORR] per RECIST v1.1) of PF-06873600 alone and in combination with letrozole or fulvestrant.

2.2 Background & Rationale To address the acquired resistance to endocrine therapy (ET) plus CDK4/6 inhibitors in HR+/HER2- breast cancer. Resistance is often driven by cyclin E overexpression and CDK2 activation. The simultaneous inhibition of CDK2/4/6 with the first-in-class agent PF-06873600 is hypothesized to overcome this intrinsic or acquired resistance and prolong clinical benefit in affected patients, as well as in patients with triple-negative breast cancer (TNBC) and ovarian cancer.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Non-randomized) **Blinding:** Open-label **Randomization:** N/A **Management Pathway:** Dose escalation followed by dose expansion with endocrine therapy.

3.2 Planned Enrollment & Duration Target \$N\$: 151 participants (Part 1 Dose Escalation \$n=78\$; Part 2 Dose Expansion \$n=73\$) **Number of Sites:** Multi-center **Estimated Study Start:** 07-Mar-2018 **Estimated Primary Completion:** 05-Apr-2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically confirmed diagnosis of advanced or metastatic disease, specifically Hormone Receptor positive (HR+) / HER2-negative breast cancer, triple-negative breast cancer (TNBC) with ≤ 2 prior lines of chemotherapy, or platinum-resistant ovarian/fallopian tube/primary peritoneal cancer with ≤ 3 prior lines of treatment. 3. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1.

4.2 Exclusion Criteria 1. Known active, uncontrolled, or symptomatic Central Nervous System (CNS) metastases. 2. Prior intolerance to letrozole or fulvestrant (for combination therapy arms). 3. Concurrent participation in another therapeutic clinical trial.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Medical Methodology: Targeted Therapy (Selective CDK2/4/6 Inhibition). **Dosage/Frequency:** Sequential escalating dose levels of PF-06873600 (1 to 50 mg) 1 to 2 times a day. The Recommended Dose for Expansion (RDE) was determined to be 25 mg twice daily (BID). **Route of Administration:** Oral. **Duration of Treatment:** Administered in 28-day cycles until disease progression, unacceptable toxicity requiring discontinuation, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Endocrine Therapy (Letrozole or Fulvestrant) in combination arms (Part 1B and Part 2). **Prohibited:** Strong or moderate inducers/inhibitors of relevant metabolizing enzymes that might interact with PF-06873600.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, Tumor Baseline Assessment
Baseline	Day 0	First Dose (Monotherapy or Combo), PK/PD baseline sampling
Interim	Weekly / Q4W	Clinic visits onsite once a week for the first 2 cycles, and every 28-day cycle (Q4W) thereafter for Safety Labs, PK draws, Efficacy Assessment
End of Study	Variable	Treatment Discontinuation, Final Vital Signs, RECIST Evaluation, AE follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of Dose-Limiting Toxicities (DLTs) in the first 28-day cycle and Objective Response Rate (ORR) defined as percentage of participants with confirmed complete response (CR) or partial response (PR).

Measurement Method: Clinical toxicity grading and Response Evaluation Criteria in Solid Tumors (RECIST) v1.1 via imaging.

7.2 Secondary & Exploratory Endpoints - Pharmacokinetics: Area under the curve ($SAUC_{ss,\tau}$) and trough concentration ($C_{ss,min}$). - Pharmacodynamics: Modulation of cell cycle PD biomarkers, specifically reductions in Ki-67 positive cells, phosphorylated Rb (pRb) histo-score, and ctDNA levels in paired tumor biopsies.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Tracked via clinic visits and laboratory tests. The most common all-causality AEs noted were nausea, anemia, fatigue, and neutropenia. **Serious Adverse Events (SAEs):** Defined as events that are fatal/life-threatening, require inpatient hospitalization, or result in persistent disability; reported promptly. **Data Monitoring Committee (DMC):** Utilized a modified toxicity probability interval method to monitor dose escalation safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set includes all enrolled patients who received at least one dose of the investigational product. **Sample Size Justification:** Determined by the modified toxicity probability interval method for dose escalation.

Handling of Missing Data: Standard handling for oncology Phase I/II trials, typically avoiding imputation for primary safety analysis.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP), the Declaration of Helsinki, and CONSORT guidelines. **Education Protocols:** Mandates investigator training on DLT definitions and PK/PD biopsy requirements (HCP Education), and onsite training for oral drug administration and symptom reporting (Patient Education). **Monitoring Plan:** High frequency of onsite monitoring, especially weekly during the first 2 cycles. Remote monitoring frequency is conducted per site capability and includes post-treatment survival follow-up. A formalized checklist review interval occurs at the end of every 28-day

cycle. **Quality Audit Mechanism & Audit Trail:** Standard strict eCRF logging, tracking dose modifications, delays, and drug accountability. Incorporates strict RECIST v1.1 independent review and real-time safety tracking.

11. Study Identifiers & Governance

Primary ID: {{C3661001}} **Registry Number:** {{NCT03519178}}
Sponsor/Lead Organization: {{Pfizer Inc.}} **Study Title:** {{A Study of PF-06873600 in People With Cancer}} **Official Regulatory Title:** {{PHASE 1/2A DOSE ESCALATION AND EXPANSION STUDY EVALUATING SAFETY, TOLERABILITY, PHARMACOKINETIC, PHARMACODYNAMICS AND ANTI-TUMOR ACTIVITY OF PF-06873600 AS A SINGLE AGENT AND IN COMBINATION WITH ENDOCRINE THERAPY}}

12. Clinical Framework (Oncology / CDK2/4/6 Specific)

Primary Condition: {{HR+ HER2- Metastatic Breast Cancer, TNBC, Ovarian Cancer}} **Diagnosis Threshold:** {{Histologically confirmed advanced or metastatic disease}} **Medical Methodology:** {{Targeted Therapy (Selective CDK2/4/6 Inhibition)}} **Optimization Target:** {{Objective Tumor Response and Prolonged Disease Control}}

13. Study Procedures & Methodology

Management Pathway: {{Dose escalation followed by dose expansion with endocrine therapy}} **Education Protocol (HCP):** {{Investigator training on DLT definitions and PK/PD biopsy requirements}} **Education Protocol (Patient):** {{Onsite training for oral drug administration and symptom reporting}} **Quality Audit Mechanism:** {{Strict RECIST v1.1 independent review and real-time safety tracking}}

14. Observation & Follow-Up Schedule

Total Duration: {{Treatment until disease progression or unacceptable toxicity}} **Onsite Visit Cadence:** {{Weekly for first 2 cycles, then every 28 days (Q4W)}} **Remote Monitoring Frequency:** {{Per site capability and post-treatment survival follow-up}} **Checklist Review Interval:** {{End of every 28-day cycle}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				
Statistician	[Name]	_____	[DD-MMM-YYYY]				